

## Capstone Project Reflection

### **Initial Codebase**

The codebase for this project was built entirely from scratch. There was no single public library or prior assignment that served as the foundation. The decision to build from scratch was deliberate rather than incidental. In sequential Bayesian optimisation, the surrogate model, the acquisition function, the candidate generation strategy, and the diagnostic pipeline need to be tightly integrated and adjusted function by function. Importing a general-purpose BO library such as BoTorch or Ax would have abstracted away precisely the decisions that mattered most, making it harder to diagnose failures, adapt hyperparameters, and understand why a particular configuration succeeded or failed. Every week's submission was accompanied by a pre-committed strategy document stating the hypothesis, the surrogate configuration, the acquisition function, and the exact conditions under which the approach would be considered falsified. Building from scratch made that level of transparency and control achievable. The full codebase, submission logs, model cards, and datasheet are publicly available at <https://github.com/HassanKhalld/Capstone-Project> [Links to an external site.](#)

### **Code Modification Week by Week**

The code evolved in six broadly recognisable phases, each driven by evidence from the previous week rather than theoretical preference. Weeks 1 and 2 established baselines using uniform random sampling and standard GP-UCB with default hyperparameters across all eight functions. This was not naivety but necessity: without prior observations, any surrogate is speculative. The goal at this stage was to get data on the table and begin EDA using correlation, mutual information, and partial correlation to identify structural signals before committing to function-specific strategies.

From Week 3 onward, each function received its own surrogate configuration. The most consequential early decision was applying a log-transform to f5's outputs from Week 3 onward. f5's output range spanned four orders of magnitude, making GP fitting in the raw scale numerically unreliable. The log-transform resolved this immediately and remained validated through to Week 13. In contrast, the same transform applied incorrectly in Week 4 as part of a log-GP plus log-SVR ensemble with a CV-LCB selection criterion produced a badly degraded result, which was the most useful failure of the entire project. The lesson was not that log-transforms are dangerous but that transformation and acquisition selection must be treated as a joint decision rather than independent choices.

Week 4 introduced ensemble methods across several functions, combining GP with SVR, KNN, and XGBoost weighted by cross-validated inverse MSE. This had its most lasting impact on f7, where the four-model ensemble proved to be the only configuration that produced sustained improvement. The ensemble approach on f7 was retained through to Week 13 and produced the final improvement in that function.

Weeks 7 and 8 were the most experimentally ambitious. Competition-validated acquisition methods were introduced across multiple functions simultaneously: Max Value Entropy Search for f2, CMA-ES hyperparameter tuning for f1 and f4, TuRBO-M with three parallel trust regions for f7, and Differential Evolution for f6. In retrospect, this phase produced the most important negative result of the project. CMA-ES and Differential Evolution generated structurally incorrect hyperparameter basins on f6 and f7, and TuRBO-M caused a significant regression on f7 by directing the search outside the confirmed high-value basin. The diagnosis from Week 8 was that gradient-free hyperparameter tuning solves the wrong problem when the GP likelihood surface already has a clear basin near the correct configuration. The methods were not flawed in principle; they were applied in the wrong condition.

From Week 9 onward, the code shifted from acquisition-function-driven search toward geometry-first and structural methods on saturated functions. For f1, a Gaussian copula transform introduced in Week 9 produced an improvement of approximately 400 million times in a single step by resolving the near-zero output scale that had rendered GP acquisition surfaces flat for six consecutive weeks. This was the single most impactful code change of the entire project. For f5, per-dimension floor constraints were tightened progressively in the upper corner of the domain, and GP posterior mean maximisation with a SIR-based tiebreaker replaced EI in the terminal phase. For f8, inactive dimensions were fixed at their incumbent values and LogEI was applied in the three-dimensional active subspace, reducing the effective search problem from eight dimensions to three.

## **Final Result**

The score evolved strongly in the final four weeks. f1 improved by approximately eight orders of magnitude between Weeks 9 and 12 through the combination of copula warping and quadratic ridge extrapolation in log10 space. f5 recorded seven consecutive improvements from Week 7 to Week 13, finishing at 8343.24 against a starting value of 1088.86. f7 reached its new best of 2.695 in the final week by testing the negative X2 gradient direction that had been identified structurally but not tested until budget was nearly exhausted. f8 produced a new best of 9.966 on the final submission by probing slightly below the incumbent in both X1 and X3, the first improvement on that function since Week 3. The project finished 9th out of 51 participants.

If given more time or a fresh start, the most significant change would be to introduce the Gaussian copula transform on f1 in Week 3 rather than Week 9. The output scale pathology was visible from the initial data and the six weeks of wasted evaluations on f1 between Weeks 3 and 8 represent the largest recoverable loss of the project. The second change would be to apply fANOVA variance decomposition to all functions from Week 3 onward rather than introducing it selectively in later weeks. fANOVA provided the most reliable dimension importance signal across every function it was applied to and would have shortened the structural identification phase on f3 and f6 by several weeks.

## **Trade-offs and Decisions**

The most significant trade-off faced throughout the project was between methodological ambition and empirical stability. In Weeks 7 and 8, the decision was made to simultaneously introduce multiple competition-validated methods across several functions. In principle this was the correct approach: competition-winning techniques exist precisely because they handle the conditions encountered in this project. In practice, introducing multiple simultaneous changes made it impossible to attribute the resulting regressions on f6 and f7 to any single component. The failure of Differential Evolution on f6 and TuRBO-M on f7 could not be isolated from each other because they were tested at the same time. The lesson is that strategy conflation, changing more than one component per week, is the most persistent limitation of a one-observation-per-function budget. The pre-commitment framework documented in the submission logs mitigated this somewhat by forcing explicit hypotheses before results were observed, but it could not resolve the attribution problem when multiple changes coincided.

The exploration versus exploitation trade-off resolved differently per function and the most important insight was that the correct resolution is not symmetric across functions. For f5, exploitation was the correct posture from Week 7 onward because the function's monotone corner structure was confirmed and the surrogate was well-calibrated. For f1, exploration would have been the correct posture from Week 3 onward had the output scale been handled properly, but the surrogate failure disguised this. For f3, neither pure exploration nor pure exploitation succeeded, and the plateau structure of that function meant that the choice of surrogate mattered more than the acquisition strategy.

## **Learning and Application**

The most important lesson from this project is that surrogate failure and function difficulty are not the same thing. For most of the first eight weeks, the functions that appeared hardest were the ones where the surrogate was most poorly specified. f1 appeared near-impossible for six weeks because the output scale made GP posterior surfaces flat, not because the function lacked structure. f4 appeared dangerous because early GP models were overconfident about the surface beyond the plateau, not because the plateau was difficult to refine. Diagnosing whether a function is genuinely difficult or whether the surrogate is simply misconfigured is the central methodological skill in expensive black-box optimisation and it is not taught in textbooks because it requires working with small samples on unknown surfaces. This project developed that diagnostic intuition directly.

The application to quantitative finance is direct. A buy-side researcher calibrating a systematic strategy faces the same problem: each backtest is expensive, the objective function has no analytical form, and the budget for evaluation is strictly finite. The surrogate-acquisition loop developed here maps cleanly onto strategy parameter search, and the diagnostic framework built into the weekly submission logs mirrors the pre-trade analysis process in quantitative research. More specifically, the insight that inactive

dimensions should be fixed at their incumbent values once confirmed, which produced the final improvement on f8, is directly analogous to fixing non-signal parameters in a systematic strategy to avoid optimisation noise corrupting the signal dimensions.

What surprised me most looking back was how consistently the simplest validated approach outperformed the most theoretically motivated one when the sample size was small. The four-model CV-weighted ensemble on f7 outperformed TuRBO-M despite TuRBO-M being backed by a NeurIPS 2019 paper directly applicable to the conditions of that function. The quadratic ridge extrapolation on f1 outperformed every GP-based acquisition strategy despite being a two-line polynomial fit. Complexity is not a virtue under budget constraints. What matters is whether the structural assumption underlying the method matches the actual structure of the function, and that match can only be verified empirically, not assumed theoretically. That is, I think, the deepest lesson of this project and the one most directly transferable to professional practice.

### **Successful Optimisation Strategies: Reflection and Analysis**

The strategies that produced the strongest results in this project were not the most technically sophisticated ones. They were the ones that were most honest about what the data actually showed. Looking back across thirteen weeks and eight functions, three principles account for the majority of the improvement: treating each submission as a falsifiable experiment rather than a guess, diagnosing surrogate failure before assuming function difficulty, and recognising when to abandon model-guided acquisition entirely in favour of direct structural reasoning.

The single most impactful strategy change of the entire project was the introduction of the Gaussian copula transform on f1 in Week 9. For six consecutive weeks, f1 had appeared near-impossible to improve. GP acquisition surfaces were completely flat, every surrogate produced near-zero predictions regardless of input, and the function looked structurally uninformative. The correct diagnosis, which took too long to arrive at, was that the failure was not in the function but in the output representation. f1's outputs span approximately 200 orders of magnitude. Fitting a Gaussian Process directly to values ranging from 10 to the power of negative 248 to 10 to the power of negative 9 is numerically incoherent because the model has no capacity to distinguish meaningful variation from rounding error. The copula transform resolved this in one step by mapping outputs through their empirical rank to a standard normal distribution, and the Week 9 submission returned a value approximately 400 million times better than the previous best. The lesson is not specific to copula transforms. It is that before concluding a function is hard, one should verify that the surrogate is actually capable of representing it.

The second strategy that defined strong results was the pre-commitment experimental design framework. Before every submission, a hypothesis was stated, the surrogate configuration was justified, and a falsification condition was written down. This sounds like administrative overhead but it had a direct effect on performance. It forced a genuine distinction between decisions made on evidence

and decisions made on intuition, and it ensured that when results came back, they could be interpreted structurally. When f7's TuRBO-M submission in Week 7 produced a severe regression, the pre-committed hypothesis made it immediately clear that the failure was caused by the trust region leaving the confirmed high-value basin, not by the ensemble weights or the acquisition beta. Without that specificity, the following week's correction would have been less targeted. The pre-commitment framework is borrowed directly from quantitative research practice in finance, where a strategy that cannot be falsified cannot be improved, and applying it to Bayesian optimisation made every failure informative rather than merely disappointing.

The third defining strategy was recognising when to stop using surrogate-guided acquisition and switch to direct geometry. For f1, a degree-two polynomial fitted in  $\log_{10}$  space to the three most locally informative observations provided a more reliable peak estimate than any GP acquisition configuration attempted. For f5, GP posterior mean maximisation with per-dimension floor constraints progressively tightened toward the upper corner produced seven consecutive improvements in the final seven weeks. For f4, an empirical floor on X3 derived directly from observed failures prevented the acquisition function from returning to a region that had already been confirmed catastrophic. In each case, the geometric or empirical constraint was more reliable than the surrogate because the surrogate was operating near the limits of what N equals approximately 30 observations can support. Knowing when a model has saturated its usefulness is at least as important as knowing how to build a better one.

Thinking about what defines a successful strategy more broadly, I would argue it is not outcomes alone. Finishing 9th out of 51 participants is a meaningful outcome, but the more durable measure of success in this project was whether each decision was made on defensible grounds. Two functions, f3 and f6, produced regressions in the final week despite using strategies that were well-justified at the time. Those regressions do not represent strategic failures in the same sense that Week 7's TuRBO-M regression on f7 did, where the failure mode was identifiable and avoidable. A strategy is successful when it extracts the maximum information from each expensive evaluation, makes the reasoning behind the decision transparent enough to be corrected, and does not confuse complexity for capability. By that measure, the quadratic ridge extrapolation on f1 was as successful a strategy as the four-model CV-weighted ensemble on f7, even though one is two lines of polynomial arithmetic and the other is a statistically weighted combination of four machine learning models.

Reflecting on peers' approaches, the most interesting contrast observed in reviewing the leaderboard and discussion posts was between participants who committed to a single surrogate family throughout the competition and those who adapted their surrogate architecture week by week. Several high-performing participants in the top ten appear to have converged early on a well-specified GP with appropriate kernel bounds and then focused their remaining budget on acquisition function tuning and candidate generation

design rather than rebuilding the surrogate from scratch each week. This is a legitimate and arguably more efficient approach than the one taken here, which was to introduce structural complexity early and then roll back when complexity failed. The advantage of the simpler approach is that rollback is never needed. The disadvantage is that it may miss structural signals that only become apparent through deliberate model diversity, such as the fANOVA variance decomposition results that identified X3 as the dominant driver of f5 or the ARD length scale analysis that confirmed X1, X3, and X7 as the active dimensions of f8.

The overlap between my approach and some peers' strategies is most visible in the use of trust-region exploitation in the final weeks. Several participants appear to have converged on local exploitation near the incumbent as the dominant strategy from around Week 9 or 10 onward, which matches the shift in this project from global exploration to tight neighbourhood search. The difference is that trust-region exploitation here was explicitly conditioned on structural diagnostics confirming the basin, whereas some approaches appear to have defaulted to local search earlier without that confirmation. The risk of the latter is visible in the f4 trajectory of this project, where early trust-region restriction before the mid-basin structure was understood contributed to slower improvement than was achievable.

One suggestion that would strengthen any approach in a future iteration of this kind of challenge is to invest more heavily in EDA in the first two to three weeks before committing to function-specific surrogate configurations. The structural signals that ultimately guided the most effective strategies, fANOVA, partial correlation, Spearman jackknife stability, and ARD length scale diagnostics, were all available from the data accumulated by Week 4 but were not fully operationalised until Weeks 6 through 8 in most cases. Running a comprehensive EDA battery immediately after Week 3 or 4 and using the results to assign each function to a structural category, monotone corner, sparse active subspace, flat plateau, narrow ridge, isolated basin, would have allowed the acquisition strategy to be matched to the structural type from the outset rather than discovered iteratively through failure.

In professional ML and AI contexts beyond this competition, the transferable lessons are specific and practical. The first is that output representation is a modelling decision as consequential as kernel choice or acquisition function selection, and it should be treated as such from the start. The second is that the exploration versus exploitation trade-off resolves differently per problem type and the correct resolution cannot be read off from a general principle. It requires structural diagnosis. The third is that a strategy that can be written down before the result is observed, and updated with specific reasoning after, is strictly more valuable than one that is adjusted in response to the outcome. This applies to strategy calibration in quantitative finance, hyperparameter search in deep learning, clinical trial design in drug development, and any other domain where evaluation is expensive and irreversible. The BBO capstone project simulates all of those conditions simultaneously and the competency it develops, making principled

decisions under uncertainty with limited budget, is not specific to optimisation. It is the central skill of applied machine learning in practice.